

Visualization Literacy Analysis

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11/25/2025

Read in data files

Raw data is in the ../AccData subdirectory in two excel spreadsheet files per participant. The following code combines into single dataframe of all data.

```
#grab files from google drive (only have to do this once)
# source("getFromGoogleDrive.R")

#get the first 6? characters of each data file
#get unique values of these
# this is list of subject ids

raw_file_names <- list.files("../AccData")
first_six <- substr(raw_file_names, 1, 6)
sub_ids <- unique(first_six)

# fast_RT = data.frame(ParticipantId = character(),
#                        TrialName = character(),
#                        type = character(),
#                        time = numeric()
# )

all_data <- NULL

for (i in 1:length(sub_ids)){

  if (grepl("~$", sub_ids[i], fixed = T)){
    next
  }

  temp_file1 <- read_xlsx(paste0("../AccData/", sub_ids[i], "_1.xlsx")) %>%
    slice(1:17) %>%
    select(-starts_with("Order")) %>%
    rename(correct = 8)

  temp_file2 <- read_xlsx(paste0("../AccData/", sub_ids[i], "_2.xlsx")) %>%
    slice(1:17) %>%
    select(-starts_with("Order")) %>%
    rename(correct = 8)

  new_temp <- temp_file1 %>%
    bind_cols(temp_file2$correct) %>%
```

```

  rename(Correct_1 = correct, Correct_2 = "...9") %>%
  mutate(AnswerRT = TimeToBeginInput - TimeToReadQuestion)

all_data <- all_data %>%
  bind_rows(new_temp)
}

all_data <- all_data %>%
  rename(readRT = TimeToReadQuestion, totalRT = TimeToBeginInput)

#read in trialtype key (I created this from an early version of the previous paper)
trial_type_key <- read.csv("../trial_type_key.csv", stringsAsFactors = F)

all_data <- all_data %>%
  mutate(TrialType = trial_type_key$TrialType[match(TrialName, trial_type_key$TrialName)]) %>%
  mutate(TrialType = paste0("Type", TrialType))

# remove temporary dataframes to keep workspace cleaner as we go forward
rm(new_temp)
rm(temp_file1)
rm(temp_file2)
rm(trial_type_key)

```

Basic checks

Just a sanity check of all the data. Summary of number of subjects in each of the 3 condition types.

```

#how many participants per condition
all_data %>%
  group_by(ParticipantId, Condition) %>%
  summarize(ntrials = n()) %>%
  group_by(Condition) %>%
  summarize(nsubs = n()) %>%
  summary()

```

`summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups` argument.

```

##   Condition      nsubs
## Length:3      Min.   :33.00
## Class :character 1st Qu.:36.00
## Mode  :character Median :39.00
##                Mean   :40.67
##                3rd Qu.:44.50
##                Max.   :50.00

```

#why is the balance so off?

Remove outliers based on RT

Removing outliers, resulting in data frame named `all_data_no_outliers`.

1. All trials where the **AnswerRT** is less than 2 seconds are removed first, which assumes these reactions were too fast so were done without an actual considered response.
2. All trials where the **AnswerRT** is 3 standard deviations different from the mean are removed next.

Table shows summary of mean and standard deviation of answer RT by each of the trial question types, as well as the 3 standard deviation upper and lower bound of the reaction time.

Displayed histogram is a sanity check of the number of trials each participant in all of the data have done. Most participants should do 16 trials. Less than that are from dropped data for outliers.

```
#removing on trial-by-trial basis

#remove answerRTs below 2000ms first
all_data_remove <- all_data %>%
  filter(AnswerRT >= 2000)
dim(all_data)[1]

## [1] 2074
dim(all_data_remove)[1]

## [1] 2067
#drops 7 trials

# also remove AnswerRT that are more than 3 standard deviations away from
# the mean
all_data_no_outliers <- all_data_remove %>%
  group_by(TrialName) %>%
  filter(!(abs(AnswerRT - mean(AnswerRT)) > 3.0*sd(AnswerRT)))
dim(all_data_no_outliers)[1]

## [1] 2020
#drops 47 more trials

rt_data_summary <- all_data_no_outliers %>%
  group_by(TrialName) %>%
  summarize(meanAnswerRT = mean(AnswerRT, na.rm = T),
            sdAnswerRT = sd(AnswerRT, na.rm = T),
            LB = meanAnswerRT - 3*sdAnswerRT,
            UB = meanAnswerRT + 3*sdAnswerRT)
summary(rt_data_summary)

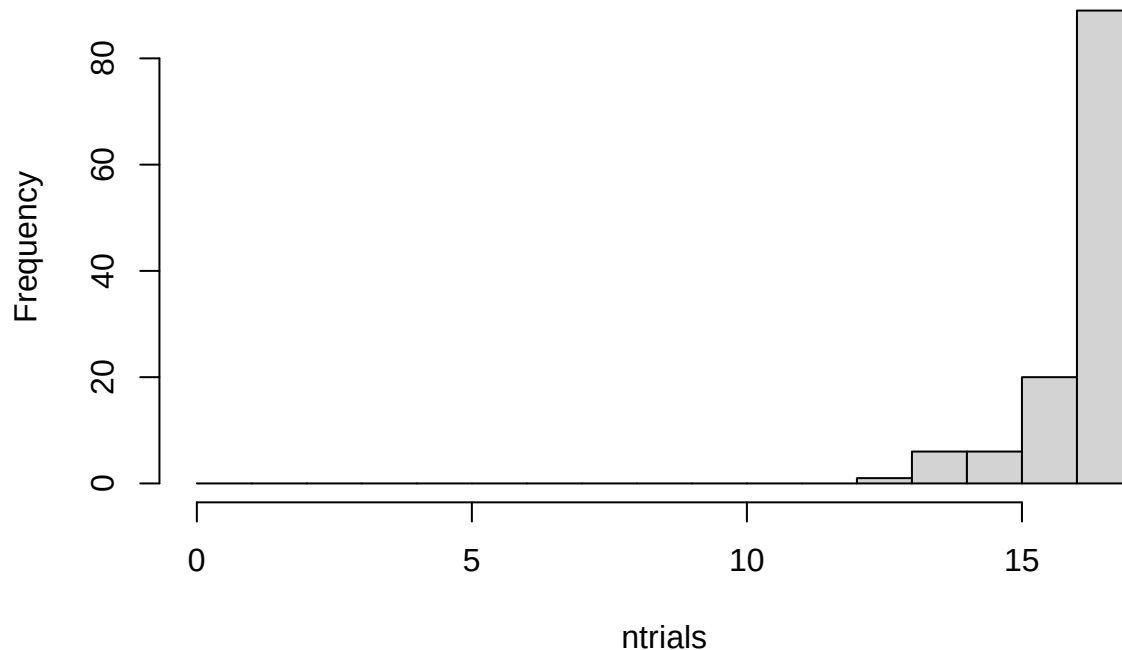
##   TrialName      meanAnswerRT    sdAnswerRT      LB      UB
## Length:17      Min.   : 8963      Min.   : 5156      Min.   :-67692      Min.   : 24429
## Class :character 1st Qu.:25275      1st Qu.:13263      1st Qu.: -35148      1st Qu.: 65715
## Mode  :character Median :29913      Median :19809      Median : -26012      Median : 88817
##              Mean  :33477      Mean  :20886      Mean  : -29182      Mean  : 96137
##              3rd Qu.:44519      3rd Qu.:26920      3rd Qu.: -17508      3rd Qu.:123649
##              Max.  :61772      Max.  :43154      Max.   : -6504      Max.   :191235

rm(rt_data_summary)

all_data_no_outliers %>%
  group_by(ParticipantId) %>%
```

```
summarize(ntrials = n()) %>%
with(hist(ntrials, breaks = 0:17))
```

Histogram of ntrials



```
rm(all_data_remove)
##maybe consider replacing outliers with means instead of removing them?
```

Extract Only One Type of Chart

Create a dataframe from the data with no outliers that contains only XChartQ1 through XChartQ4 trials for the following analysis.

We replace the `all_data_no_outliers` dataframe with one containing only the selected TrialName question type here, so past this point we are analyzing only this particular TrialName type.

We show the table again of the mean RT for this question type. The table should match the one done with all of the data, but we should only have the particular question trial name type now in the data.

We also again plot the histogram of the number of trials. Participants should have seen 4 of each particular question type, but after removing outliers some only have 3, 2 or 1 remaining.

```
all_data_no_outliers <- all_data_no_outliers %>%
  filter(grepl("LineChartQ[0-9]", TrialName))

rt_data_summary <- all_data_no_outliers %>%
  group_by(TrialName) %>%
  summarize(meanAnswerRT = mean(AnswerRT, na.rm = T),
            sdAnswerRT = sd(AnswerRT, na.rm = T),
            LB = meanAnswerRT - 3*sdAnswerRT,
```

```

UB = meanAnswerRT + 3*sdAnswerRT
summary(rt_data_summary)

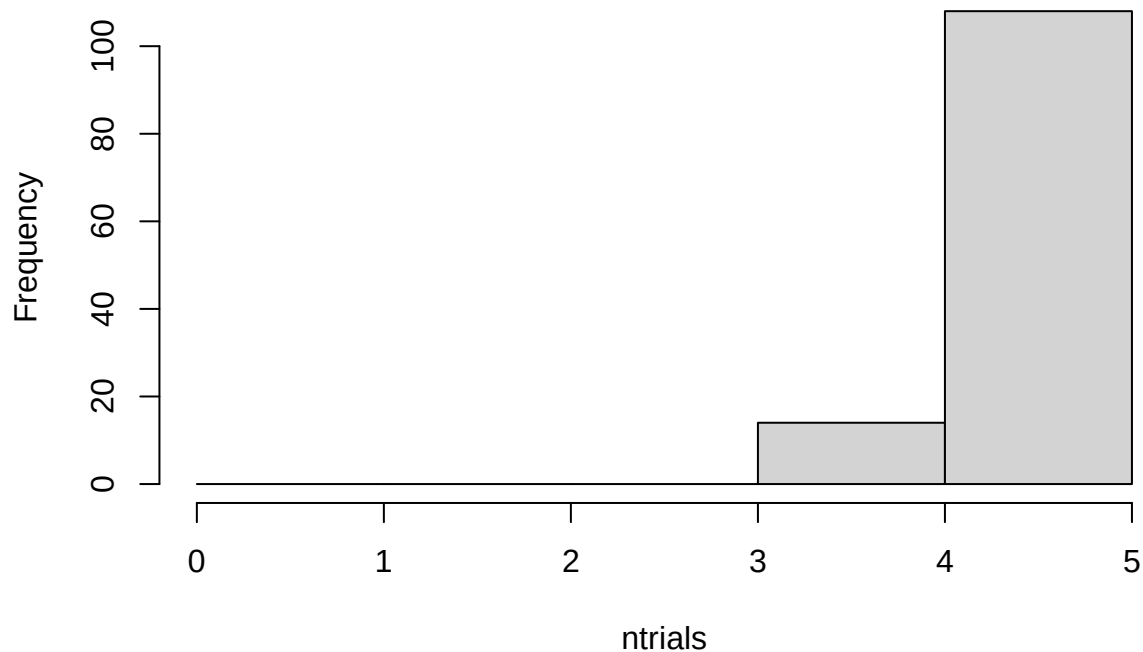
##   TrialName      meanAnswerRT    sdAnswerRT      LB      UB
## Length:5      Min.   :13372    Min.   :13263    Min.   : -33739    Min.   : 53161
## Class :character 1st Qu.:25275    1st Qu.:14261    1st Qu.: -26416    1st Qu.: 68057
## Mode  :character Median :27265    Median :15938    Median : -24437    Median : 75080
##              Mean   :30097    Mean   :18209    Mean   : -24530    Mean   : 84724
##              3rd Qu.:38246    3rd Qu.:23588    3rd Qu.: -20549    3rd Qu.:110232
##              Max.   :46327    Max.   :23995    Max.   : -17508    Max.   :117090

rm(rt_data_summary)

all_data_no_outliers %>%
  group_by(ParticipantId) %>%
  summarize(ntrials = n()) %>%
  with(hist(ntrials, breaks = 0:5))

```

Histogram of ntrials



Compare Answer RT

```

#compare answer times

trial_type_means <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition, TrialType) %>%
  summarize(mean_answerRT = mean(AnswerRT), n = n())

```

```
## `summarise()` has grouped output by 'ParticipantId', 'Condition'. You can override using
## the `.groups` argument.
```

```
#rm(trial_type_means)
```

Alternative Analysis using glmm TMB package

The following analysis is for the heterogenous group condition analysis. See .

The following determines if a more complex model is justified.

```
model_glmm_hom = glmmTMB(mean_answerRT ~ Condition * TrialType + (1 | ParticipantId),
                          data=trial_type_means, family = Gamma(link="log"),
                          dispformula = ~ 1, REML = FALSE)
model_glmm_hetero = glmmTMB(mean_answerRT ~ Condition * TrialType + (1 | ParticipantId),
                             data=trial_type_means, family = Gamma(link="log"),
                             dispformula = ~ Condition, REML = FALSE)

anova(model_glmm_hom, model_glmm_hetero)
```

```
## Data: trial_type_means
```

```
## Models:
```

```
## model_glmm_hom: mean_answerRT ~ Condition * TrialType + (1 | ParticipantId), zi=~0, disp=~1
```

```
## model_glmm_hetero: mean_answerRT ~ Condition * TrialType + (1 | ParticipantId), zi=~0, disp=~Condition
```

```
##           Df    AIC    BIC logLik deviance Chisq Chi Df Pr(>Chisq)
```

```
## model_glmm_hom      11 7866.9 7909.7 -3922.4   7844.9
```

```
## model_glmm_hetero  13 7870.6 7921.2 -3922.3   7844.6 0.2334      2    0.8898
```

Try overall effect reporting

```
fixed_effects_anova <- Anova(model_glmm_hom, type="III")
kable(fixed_effects_anova)
```

	Chisq	Df	Pr(>Chisq)
(Intercept)	20119.623676	1	0.0000000
Condition	8.015497	2	0.0181743
TrialType	49.013210	2	0.0000000
Condition:TrialType	9.708390	4	0.0456370

```
#summary(fixed_effects_anova, level=0.95)
```

EMM by VR, VR Monitor, VR Monitor Stereo

```
emm_main_condition <- emmeans(model_glmm_hom,
                               specs = ~ Condition,
                               type = "response")
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```
summary(emm_main_condition, level=0.95)
```

```
## Condition      response    SE  df asymp.LCL asymp.UCL
## VR              25709 1450 Inf    23016    28718
## VR Monitor      28878 1830 Inf    25500    32704
```

```
## VR Monitor Stereo      25997 1790 Inf      22709      29760
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
```

```
confint(emm_main_condition, level=0.95)
```

```
## Condition      response    SE  df asymp.LCL asymp.UCL
## VR              25709 1450 Inf      23016      28718
## VR Monitor      28878 1830 Inf      25500      32704
## VR Monitor Stereo 25997 1790 Inf      22709      29760
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
```

Preregistered contrasts.

```
comparison_results <- contrast(
  emm_main_condition,
  method = list(
    VR_vs_Monitors_Collapsed = c('VR' = 1, 'VR Monitor' = -0.5, 'VR Monitor Stereo' = -0.5)
  )
)
```

```
summary(comparison_results, infer = TRUE)
```

```
## contrast          ratio      SE  df asymp.LCL asymp.UCL null z.ratio p.value
## VR_vs_Monitors_Collapsed 0.938 0.0688 Inf      0.813      1.08      1 -0.869 0.3850
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
## Tests are performed on the log scale
```

```
confint(comparison_results, level = 0.95)
```

```
## contrast          ratio      SE  df asymp.LCL asymp.UCL
## VR_vs_Monitors_Collapsed 0.938 0.0688 Inf      0.813      1.08
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
```

Preregistered contrasts.

```
comparison_results <- contrast(
  emm_main_condition,
  method = list(
    VRMonitor_vs_others_collapsed = c('VR' = -0.5, 'VR Monitor' = 1, 'VR Monitor Stereo' = -0.5)
  )
)
```

```
summary(comparison_results, infer = TRUE)
```

```
## contrast          ratio      SE  df asymp.LCL asymp.UCL null z.ratio p.value
## VRMonitor_vs_others_collapsed 1.12 0.0866 Inf      0.96      1.3      1 1.428 0.1533
```

```
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
## Tests are performed on the log scale
confint(comparison_results, level = 0.95)

## contrast          ratio      SE  df asymp.LCL asymp.UCL
## VRMonitor_vs_others_collapsed  1.12 0.0866 Inf      0.96      1.3
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
```

EMM on Condition and Interaction

If the glmm analysis is not significant, just use the homogeneous model. Otherwise report on the more complex heterogeneous model?

```
emm_main_condition <- emmeans(model_glmm_hom,
                              specs = ~ Condition,
                              type = "response")

## NOTE: Results may be misleading due to involvement in interactions

condition_results <- pairs(emm_main_condition)

summary(condition_results, level=0.95)

## contrast          ratio      SE  df null z.ratio p.value
## VR / VR Monitor      0.890 0.0755 Inf    1  -1.370  0.3569
## VR / VR Monitor Stereo  0.989 0.0881 Inf    1  -0.125  0.9914
## VR Monitor / VR Monitor Stereo 1.111 0.1040 Inf    1   1.122  0.5004
##
## Results are averaged over the levels of: TrialType
## P value adjustment: tukey method for comparing a family of 3 estimates
## Tests are performed on the log scale
confint(condition_results, level=0.95)

## contrast          ratio      SE  df asymp.LCL asymp.UCL
## VR / VR Monitor      0.890 0.0755 Inf    0.730    1.09
## VR / VR Monitor Stereo  0.989 0.0881 Inf    0.803    1.22
## VR Monitor / VR Monitor Stereo 1.111 0.1040 Inf    0.892    1.38
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Conf-level adjustment: tukey method for comparing a family of 3 estimates
## Intervals are back-transformed from the log scale
#kable(condition_results)

emm_interaction <- emmeans(model_glmm_hom,
                           specs = ~ Condition * TrialType,
                           type = "response")
```

```
simple_effects_results <- pairs(emm_interaction, by = "TrialType")
```

```
summary(simple_effects_results, level=0.95)
```

```
## TrialType = Type1:
## contrast ratio SE df null z.ratio p.value
## VR / VR Monitor 0.729 0.0814 Inf 1 -2.831 0.0129
## VR / VR Monitor Stereo 0.867 0.1020 Inf 1 -1.211 0.4465
## VR Monitor / VR Monitor Stereo 1.190 0.1480 Inf 1 1.404 0.3386
##
## TrialType = Type2:
## contrast ratio SE df null z.ratio p.value
## VR / VR Monitor 0.937 0.1070 Inf 1 -0.568 0.8374
## VR / VR Monitor Stereo 0.966 0.1150 Inf 1 -0.290 0.9546
## VR Monitor / VR Monitor Stereo 1.030 0.1280 Inf 1 0.241 0.9685
##
## TrialType = Type3:
## contrast ratio SE df null z.ratio p.value
## VR / VR Monitor 1.033 0.1160 Inf 1 0.288 0.9553
## VR / VR Monitor Stereo 1.154 0.1360 Inf 1 1.221 0.4407
## VR Monitor / VR Monitor Stereo 1.118 0.1390 Inf 1 0.897 0.6421
##
## P value adjustment: tukey method for comparing a family of 3 estimates
## Tests are performed on the log scale
```

```
confint(simple_effects_results, level=0.95)
```

```
## TrialType = Type1:
## contrast ratio SE df asymp.LCL asymp.UCL
## VR / VR Monitor 0.729 0.0814 Inf 0.561 0.947
## VR / VR Monitor Stereo 0.867 0.1020 Inf 0.659 1.142
## VR Monitor / VR Monitor Stereo 1.190 0.1480 Inf 0.890 1.591
##
## TrialType = Type2:
## contrast ratio SE df asymp.LCL asymp.UCL
## VR / VR Monitor 0.937 0.1070 Inf 0.718 1.224
## VR / VR Monitor Stereo 0.966 0.1150 Inf 0.730 1.278
## VR Monitor / VR Monitor Stereo 1.030 0.1280 Inf 0.770 1.378
##
## TrialType = Type3:
## contrast ratio SE df asymp.LCL asymp.UCL
## VR / VR Monitor 1.033 0.1160 Inf 0.794 1.344
## VR / VR Monitor Stereo 1.154 0.1360 Inf 0.876 1.521
## VR Monitor / VR Monitor Stereo 1.118 0.1390 Inf 0.836 1.495
##
## Confidence level used: 0.95
## Conf-level adjustment: tukey method for comparing a family of 3 estimates
## Intervals are back-transformed from the log scale
```

```
#kable(simple_effects_results)
```

Compare Accuracy

Extract wide dataframe

Do all of the previous again but on the Accuracy results.

We again create a dataframe in needed format of only the accuracy data.

A bit different from before, only need to create an average of the two correct features we have. But then we do a similar transformation into wide format and replace na/missing values.

```
all_data_no_outliers <- all_data_no_outliers %>%  
  rowwise() %>%  
  mutate(Correct_Avg = mean(c(Correct_1, Correct_2)))
```

```
trial_type_mean_accuracy <- all_data_no_outliers %>%  
  group_by(ParticipantId, Condition, TrialType) %>%  
  summarize(mean_acc = mean(Correct_Avg),  
            n = n())
```

`summarise()` has grouped output by 'ParticipantId', 'Condition'. You can override using
the `groups` argument.

```
table(all_data_no_outliers$Correct_Avg)
```

```
##  
##    0 0.25  0.5 0.75    1  
## 196    1   46    1 352
```

```
#rm(trial_type_mean_accuracy)
```

```
# Step 1: Aggregate to the Participant Level  
# In a mixed design, the experimental unit is the Participant.  
# We first average all raw trials for each person within each Trial Type.
```

```
subject_means <- all_data_no_outliers %>%  
  group_by(ParticipantId, Condition, TrialType) %>%  
  summarize(  
    subject_avg = mean(Correct_Avg, na.rm = TRUE),  
    .groups = "drop"  
  )
```

```
# Step 2: Compute Descriptive Stats for the Groups  
# Now we calculate the stats across participants.
```

```
descriptive_stats <- subject_means %>%  
  group_by(Condition) %>%  
  summarize(  
    n = n(), # Number of subjects per group  
    Mean = mean(subject_avg), # Grand Mean  
    SD = sd(subject_avg), # Standard Deviation (Variability)  
    SE = SD / sqrt(n), # Standard Error
```

```
# 95% Confidence Interval Calculation  
# We use qt(0.975) for the two-tailed t-distribution critical value  
CI_Margin = qt(0.975, df = n - 1) * SE,  
CI_Lower = Mean - CI_Margin,  
CI_Upper = Mean + CI_Margin,  
  .groups = "drop"  
)
```

```

# View the final table
print(descriptive_stats)

## # A tibble: 3 x 8
##   Condition      n Mean   SD   SE CI_Margin CI_Lower CI_Upper
##   <chr>      <int> <dbl> <dbl> <dbl>   <dbl>   <dbl>   <dbl>
## 1 VR          146 0.654 0.386 0.0320  0.0632   0.591   0.717
## 2 VR Monitor   117 0.560 0.411 0.0380  0.0753   0.485   0.635
## 3 VR Monitor Stereo 99 0.604 0.398 0.0400  0.0793   0.524   0.683

# Step 2: Compute Descriptive Stats for the Groups
# Now we calculate the stats across participants.
descriptive_stats <- subject_means %>%
  group_by(TrialType) %>%
  summarize(
    n = n(),                                # Number of subjects per group
    Mean = mean(subject_avg),               # Grand Mean
    SD = sd(subject_avg),                  # Standard Deviation (Variability)
    SE = SD / sqrt(n),                    # Standard Error

    # 95% Confidence Interval Calculation
    # We use qt(0.975) for the two-tailed t-distribution critical value
    CI_Margin = qt(0.975, df = n - 1) * SE,
    CI_Lower = Mean - CI_Margin,
    CI_Upper = Mean + CI_Margin,
    .groups = "drop"
  )

# View the final table
print(descriptive_stats)

## # A tibble: 3 x 8
##   TrialType      n Mean   SD   SE CI_Margin CI_Lower CI_Upper
##   <chr>    <int> <dbl> <dbl> <dbl>   <dbl>   <dbl>   <dbl>
## 1 Type1    122 0.727 0.330 0.0299  0.0591   0.668   0.787
## 2 Type2    118 0.521 0.500 0.0460  0.0911   0.430   0.612
## 3 Type3    122 0.578 0.318 0.0288  0.0571   0.521   0.635

```

Binary Collapse

```

library(ordinal)

##
## Attaching package: 'ordinal'

## The following object is masked from 'package:dplyr':
##
##   slice

# Collapse to Binary: Incorrect (0) vs. Correct (1)
# Treat 0.25, 0.5, 0.75 as "partially correct" = Correct
all_data_no_outliers <- all_data_no_outliers %>%
  mutate(Correct_Binary = ifelse(Correct_Avg == 0, 0, 1),
         Correct_Binary_Factor = factor(Correct_Binary,
                                       levels = c(0, 1),
                                       labels = c("Incorrect", "Correct")),

```

```

                                ordered = TRUE))

# Check distribution
table(all_data_no_outliers$Correct_Binary_Factor)

##
## Incorrect    Correct
##          196      400

table(all_data_no_outliers$Correct_Binary_Factor,
      all_data_no_outliers$Condition)

##
##              VR VR Monitor VR Monitor Stereo
## Incorrect    73          69                54
## Correct     171         120                109

# Fit binary ordinal model (essentially logistic regression with random effects)
model_clmm_binary <- clmm(Correct_Binary_Factor ~ Condition * TrialType + (1 | ParticipantId),
                          data = all_data_no_outliers,
                          link = "logit")

summary(model_clmm_binary)

## Cumulative Link Mixed Model fitted with the Laplace approximation
##
## formula: Correct_Binary_Factor ~ Condition * TrialType + (1 | ParticipantId)
## data:    all_data_no_outliers
##
## link threshold nobs logLik AIC      niter      max.grad cond.H
## logit flexible  596  -348.60 717.20 715(1444) 1.67e-04 1.6e+02
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## ParticipantId (Intercept) 0.3138   0.5602
## Number of groups: ParticipantId 122
##
## Coefficients:
##                                Estimate Std. Error z value Pr(>|z|)
## ConditionVR Monitor            -0.0484    0.4216  -0.115  0.90862
## ConditionVR Monitor Stereo      0.3658    0.4704   0.778  0.43671
## TrialTypeType2                  -1.0194    0.4152  -2.455  0.01409 *
## TrialTypeType3                  -1.0873    0.3436  -3.164  0.00155 **
## ConditionVR Monitor:TrialTypeType2 -0.6952    0.6130  -1.134  0.25679
## ConditionVR Monitor Stereo:TrialTypeType2 -1.1402    0.6636  -1.718  0.08578 .
## ConditionVR Monitor:TrialTypeType3 -0.2514    0.5168  -0.487  0.62661
## ConditionVR Monitor Stereo:TrialTypeType3 -0.5323    0.5659  -0.941  0.34695
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Threshold coefficients:
##                                Estimate Std. Error z value
## Incorrect|Correct          -1.5960    0.2837  -5.626

```